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SVD

Positive definite matrices

↳ a matrix with only +ve eigen values.

To check:
we will rely on the pivots of the echelon form.

of +ve pivots = # of +ve λ's (/-ve/0) } only the sign, values don't tell anything.

Properties of PD matrices:

- for PD matrices → all pivots will be +ve.
- Symmetric
- only +ve λ's ⇒ all +ve pivots
- $x^T A x > 0 \quad \forall x \neq 0$

Positive semi-definite matrices:

PSD:
- Symmetric
- All λ's ≥ 0 ⇒ All pivots ≥ 0
- $x^T A x \geq 0$

Negative definite matrices (ND)

- Symmetric
- λ's < 0 ⇒ All -ve pivots
- $x^T A x < 0, \quad \forall x \neq 0$

NSD:
- Symmetric
- λ's ≤ 0 ⇒ Pivots ≤ 0
- $x^T A x \leq 0$

Theorems:

- If A is PD ⇒ A^{-1} is also PD.

Proof: eigen values for A → λ_is (all +ves)
⇒ eigen values for A^{-1} → $\frac{1}{\lambda_i}$s ⇒ all +ves as well
∴ A^{-1} is PD as well.

- Given a polynomial

$$ax_1^2 + bx_1x_2 + cx_2^2$$

we can construct only many matrices $\begin{pmatrix} a & b_1 \\ b_2 & c \end{pmatrix}$, $b_1 + b_2 = b$

For PD matrix, it has constraints:
- $b_1 = b_2$
- $P(x_1, x_2) > 0$ $\begin{pmatrix} a & b/2 \\ b/2 & c \end{pmatrix}$

- If A and B are PD/PSD, then A+B will be PD/PSD.

Proof:
$$x^T A x > 0$$
$$x^T B x > 0$$
$$x^T A x + x^T B x > 0$$
$$x^T (A+B) x > 0$$

Similar matrices

If 2 matrices are similar, $A = S^{-1}BS$ under the existance of an invertible matrix S.

$A \sim B$ iff $A = SBS^{-1}$ or $B = SAS^{-1}$

$AS = SB$ $\quad$ $BS = SA$

Alt: 2 matrices are similar iff they have the same rank and same char. polynomial (same minimal polynomial).

Properties:

- A matrix is similar to itself. $A \sim B$.
- If $A \sim B \Rightarrow B \sim A$.
- If 2 matrices are similar, then their characteristic eqns will be same.

  similarity $\Rightarrow$ same characteristic eqns (NOT the other way round)
- If $A \sim B$, $B \sim C$, then $A \sim C$.

  Proof: $B = SAS^{-1}$

  $C = DBD^{-1}$

  $= DSAS^{-1}D^{-1}$

  $= DS\ A\ (DS)^{-1}$
- Similar matrices have same eigen values.
- Similar matrices have the same rank.

  Proof: $\text{Rank}(A) + \text{Nullity}(A) = n$

  $AS = SB$

  $\text{Null}(A) = \{\vec{v} \in \mathbb{R}^n \mid A\vec{v} = \vec{0}\}$

  Lemma: If $x \in \text{Null}(B) \Rightarrow x \in \text{Null}(A)$

  Proof: $B\vec{x} = \vec{0}$

  $\Rightarrow SB\vec{x} = S\vec{0}$ or $A(S\vec{x}) = SB\vec{x} = \vec{0}$

  $\Rightarrow SB\vec{x} = \vec{0}$

  $\Rightarrow A\vec{x} = \vec{0}$

  Lemma: If $v_1, \ldots, v_n \in \text{Null}(B)$ and are a basis for Null(B), then $Sv_1, \ldots, Sv_n$ are LI and are a basis for Null(A).

  Proof:

  $c_1Sv_1 + c_2Sv_2 + \cdots + c_nSv_n = 0$

  (If Sv_is are LI then c_is are 0)

  $S(c_1v_1 + \ldots + c_nv_n) = 0$

  $S S^{-1}(c_1v_1 + \ldots + c_nv_n) = S^{-1} \cdot 0$

  $c_1v_1 + \ldots + c_nv_n = 0$

  Hence $k = \dim(\text{Null}(B)) \leq \dim(\text{Null}(A))$

  similarly if we start with $B = SAS^{-1}$

  $\Rightarrow k = \dim(\text{Null}(A)) \leq \dim(\text{Null}(B))$

  $\dim(\text{Null}(B)) = \dim(\text{Null}(A)) \Rightarrow \text{rank}(B) = \text{rank}(A)$
- characteristic eqns being same does NOT imply that the matrices are similar.
- Similar matrices have same trace and determinant.
- similar matrices (same λs) may not have same eigen vectors.
- If A and B are of the same order, AB and BA are similar matrices.

Diagonalization

Can we find a similar diagonal matrix to a given matrix A?

$$A = SDS^{-1}$$

$$Av = \lambda v$$

$$A\begin{bmatrix} | & | & & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & & | \end{bmatrix} = \begin{bmatrix} | & | & & | \\ Av_1 & Av_2 & \cdots & Av_n \\ | & | & & | \end{bmatrix} = \begin{bmatrix} | & | & & | \\ \lambda_1 v_1 & \lambda_2 v_2 & \cdots & \lambda_n v_n \\ | & | & & | \end{bmatrix} = \begin{bmatrix} | & | & & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & & | \end{bmatrix} \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & & \vdots \\ \vdots & & \ddots & 0 \\ 0 & \cdots & 0 & \lambda_n \end{bmatrix}$$

$$A = V \Lambda V^{-1}$$

Special case: real symmetric matrix

- n LI eigen vectors.
- All eigen vectors are orthogonal.

$|v| = 1$ If we convert these orthogonal vectors to orthonormal, then matrix $V \cdot V^T = I$

$$\Rightarrow V^T = V^{-1}$$

$$A = V \Lambda V^{-1} = V \Lambda V^T$$

$$A = VDV^{-1}$$

In terms of PCA,

D → sort the diagonal entries in descending order of λ's and reorder the V matrix accordingly.

$$A = VDV^T$$
$$A^2 = VD^2V^T$$
$$\vdots$$
$$A^k = VD^kV^T$$

